

**2023 Spring CH223 Organic Chemistry II. Topic list for the POST exam-3 materials. (The final exam questions will come/derive from POST exam3 materials + three mid-term exams + practice problems).**

The concepts, reaction conditions (e.g. reagents), reaction product structure, and arrow-pushing reaction mechanisms of the following topics should be mastered for the exam (unless otherwise noted). See the exam practice problem to see how those will be asked in the exam.

The topics of the previous exams will not be the main focus of this exam, but some concepts/terms/reactions of the previous topics may be involved for this exam (e.g. benzene is the 6-member ring aromatic compound which was learned in the exam1 topics).

A topic highlighted in cyan is about reaction mechanisms. The cyan highlight does NOT indicate that it is more important than other topics.

### **Class33 amine intro (Loudon 23, Carey 22)**

Amines are basic and nucleophilic.

Alkylation of amines with alkyl halide (e.g. methyl iodide).

Mechanism of alkylation of amines with alkyl halide (SN2 reactions)

Other alkylation methods with epoxide (three member ring with oxygen) and alpha,beta-unsaturated carbonyl/nitrile.

### **Class34 reductive amination (Loudon 23, Carey 22)**

Reductive amination (also called reductive alkylation) by amine and aldehyde/ketone with hydride.

Mechanism of the reductive amination (Schiff base/imine formation first, and then hydride reduction of the imine).

Hofmann elimination occurs with a quaternary ammonium salt with hydroxide ion to provide alkene.

Mechanism of Hofmann elimination.

Amines react with sodium nitrite,  $\text{NaNO}_2$  (not nitrate which is  $\text{NaNO}_3$ ) to produce diazonium salt. (the formation of the diazonium is called diazotization)

Mechanism of the formation of the diazonium salt.

Alkyl (also called aliphatic) diazonium salts releases  $\text{N}_2$  gas and generate carbocation that produces SN1 or E1 product (i.e. alcohol or alkene)

Aryl diazonium can be replaced with halide such as chloride and bromide using copper halide (Sandmeyer reaction).

Reactions of aryl diazonium with cyano, iodide,  $\text{H}_2\text{O}/\text{Cu}_2\text{O}$  (hydrolysis),  $\text{H}_2\text{PO}_2$  (replacement with hydrogen) will NOT be asked in the final exam.

### **Class35 azo compounds (Loudon 23, Carey 22)**

Reaction of diazonium with (activated) benzene derivatives produces azo compounds.

#### **Mechanism for azo compound formation.**

Gabriel synthesis to make primary amine with phthalimide from alkyl halide without secondary and tertiary amine side products

Staudinger reaction to make primary amine with sodium azide ( $\text{NaN}_3$ ) followed by phosphine ( $\text{PPh}_3$ ) from alkyl halide without secondary and tertiary amine side products.

Reduction of nitro group with  $\text{Sn}/\text{HCl}$  or  $\text{Fe}/\text{HCl}$  produces primary amine groups

### **Class36 Pd amination (Loudon 23, Carey 22)**

Buchwald-Hartwig amination is another Pd cross coupling of aryl halide but with amines (instead of boronic acid in Suzuki coupling or alkene in Heck coupling).